

Difference Between Analog and Digital signal

What is an Analog Signal?

[Analog signal](#) is a type of signal that represents continuous data using a continuous range of values. In other words, it can take on any value within a certain range. Analog signals are characterized by their smooth and continuous nature.

For example, imagine a sound wave. In analog form, the sound wave is represented by a continuously varying electrical signal that mirrors the fluctuations in air pressure caused by the sound.

What is Digital Signal?

[Digital signal](#) is a type of signal that represents data as a sequence of discrete values, typically using binary numbers, it also contain different voltage values. Unlike analog signals, which vary continuously over time, digital signals are discrete and quantized, meaning they only take on specific, distinct values.

In digital communication and computing systems, information is encoded into digital signals for transmission, processing, and storage. These signals are commonly used in telecommunications, audio and video processing, computer networks, and many other applications.

Difference Between Analog And Digital Signal

The difference between analog signal and digital signal could be understood from the table given below:

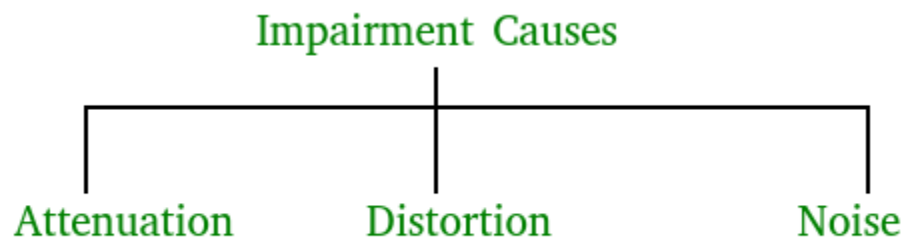
Basis	Analog Signal	Digital Signal
Definition	Analog signals represent continuous variations in magnitude over time.	Digital signals are Discrete and quantized, with specific values.
Signal Type	Continuous waveforms	Discrete Signals
Processing	Requires complex processing for manipulation.	Easier to process and manipulate digitally.

Basis	Analog Signal	Digital Signal
Storage	Less efficient for storage due to continuous nature.	More efficient for storage due to discrete values.
Bandwidth	Typically requires more bandwidth.	Requires less bandwidth for transmission.
Examples	Analog audio signals, analog radio waves, Human voice, etc.	Digital audio signals, digital data streams, computers, etc.
Errors	Susceptible to noise and distortion	More resistant to noise and distortion
Circuit Component	Amplifiers, filters, continuous-wave oscillators	Microprocessors, binary counters, logic gates
Signal Values	Infinite range of values	Limited to discrete values
Conversion	No conversion required	Analog-to-digital conversion (ADC) required
Applications	Analog signals are used in electric fan, landlines, radio frequency communications, etc.	Digital signals are used in computers, smartphones, digital sensors, digital imaging, etc.

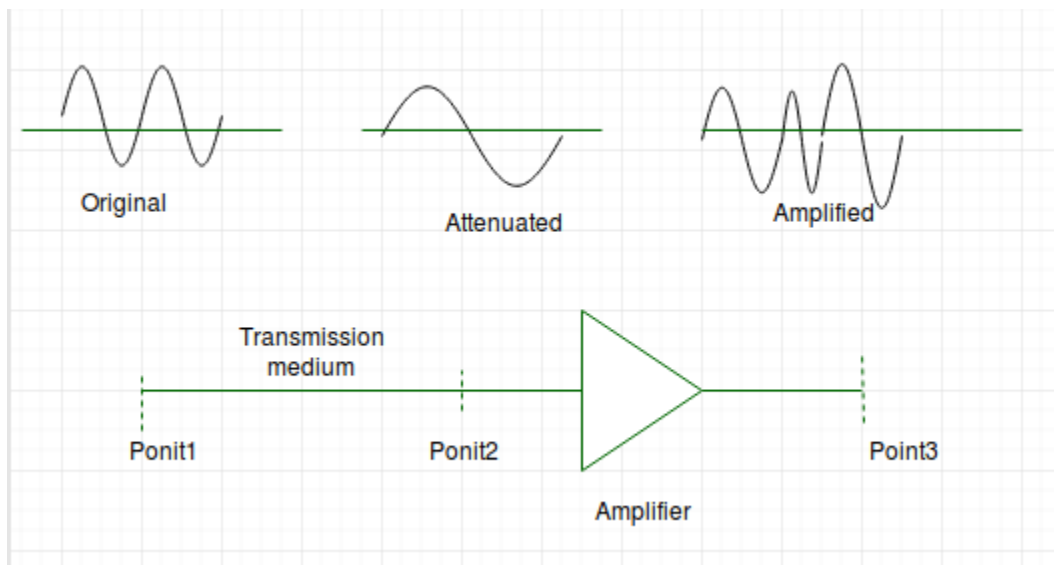
Transmission Impairment in Data Communication

In communication system, analog signals travel through transmission media, which tends to deteriorate the quality of analog signal, which means that the signal at the beginning of the medium is not the same as the signal at the end of the medium. The imperfection causes signal impairment. Below are the causes of the impairment.

Causes of impairment –



- **Attenuation** – It means loss of energy. The strength of signal decreases with increasing distance which causes loss of energy in overcoming resistance of medium. This is also known as attenuated signal. Amplifiers are used to amplify the attenuated signal which gives the original signal back and compensate for this loss.



- Image Source – aviationchief

Attenuation is measured in **decibels(dB)**. It measures the relative strengths of two signals or one signal at two different point.

$$\text{Attenuation(dB)} = 10\log_{10}(P_2/P_1)$$

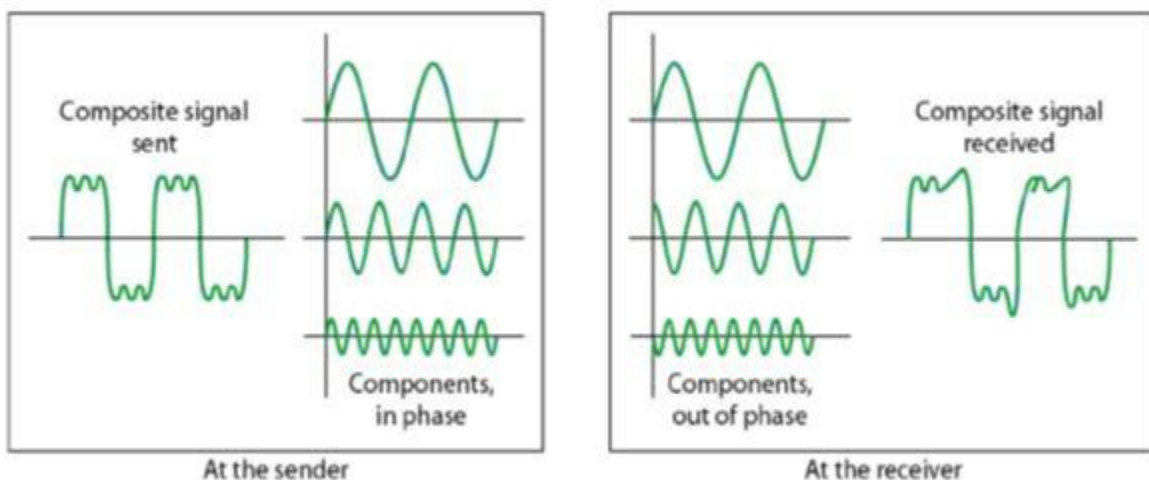
P1 is the power at sending end and P2 is the power at receiving end.

Some where the decibel is also define in terms of voltage instead of power. In this case because power is proportional to the square of the voltage the formula is

$$\text{Attenuation(dB)} = 20\log_{10}(V_2/V_1)$$

V1 is the voltage at sending end and V2 is the voltage at receiving end.

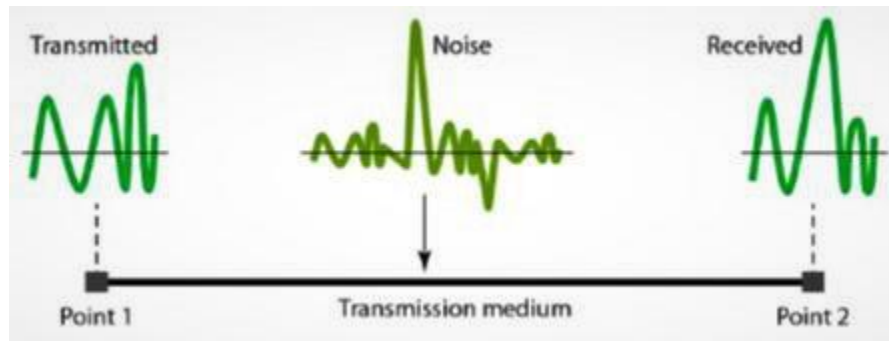
- **Distortion** – It means changes in the form or shape of the signal. This is generally seen in composite signals made up with different frequencies. Each frequency component has its own propagation speed travelling through a medium. And thats why it delay in arriving at the final destination Every component arrive at different time which leads to distortion. Therefore, they have different phases at receiver end from what they had at senders end.



- **Noise** – The random or unwanted signal that mixes up with the original signal is called noise. There are several types of noise such as induced noise, crosstalk noise, thermal noise and impulse noise which may corrupt the signal.

Induced noise comes from sources such as motors and appliances. These devices act as sending antenna and transmission medium act as receiving antenna. **Thermal** noise is movement of

electrons in wire which creates an extra signal. **Crosstalk** noise is when one wire affects the other wire. **Impulse** noise is a signal with high energy that comes from lightning or power lines



- To find the theoretical bit rate limit, we need to know the ratio. The signal-to-noise ratio is defined as

$$\text{SNR} = \text{AVG SIGNAL POWER} / \text{AVG NOISE POWER}$$

$$\text{SNR}_{\text{dB}} = 10 \log_{10} \text{SNR}$$

EXAMPLE

The values of SNR and SNR_{dB} for a noiseless channel are

$$\text{SNR} = \text{Signal Power} / 0 = \infty$$

$$\text{SNR}_{\text{dB}} = 10 \log_{10} \infty = \infty$$

We can never achieve this ratio in real life ; it is an ideal.

Performance of a Network

The performance of a network pertains to the measure of service quality of a network as perceived by the user. There are different ways to measure the performance of a network, depending upon the nature and design of the network. Finding the performance of a network depends on both quality of the network and the quantity of the network.

Parameters for Measuring Network Performance

- [Bandwidth](#)
- Latency (Delay)
- [Bandwidth – Delay Product](#)

- [Throughput](#)
- Jitter

BANDWIDTH

One of the most essential conditions of a website's performance is the amount of bandwidth allocated to the network. Bandwidth determines how rapidly the webserver is able to upload the requested information. While there are different factors to consider with respect to a site's performance, bandwidth is every now and again the restricting element.

Bandwidth is characterized as the measure of data or information that can be transmitted in a fixed measure of time. The term can be used in two different contexts with two distinctive estimating values. In the case of digital devices, the bandwidth is measured in bits per second(bps) or bytes per second. In the case of analog devices, the bandwidth is measured in cycles per second, or Hertz (Hz).

Bandwidth is only one component of what an individual sees as the speed of a network. People frequently mistake bandwidth with internet speed in light of the fact that [Internet Service Providers](#) (ISPs) tend to claim that they have a fast "40Mbps connection" in their advertising campaigns. True internet speed is actually the amount of data you receive every second and that has a lot to do with latency too. **"Bandwidth" means "Capacity" and "Speed" means "Transfer rate"**.

More bandwidth does not mean more speed. Let us take a case where we have double the width of the tap pipe, but the water rate is still the same as it was when the tap pipe was half the width. Hence, there will be no improvement in speed. When we consider [WAN](#) links, we mostly mean bandwidth but when we consider LAN, we mostly mean speed. This is on the grounds that we are generally constrained by expensive cable bandwidth over WAN rather than hardware and interface data transfer rates (or speed) over [LAN](#).

- **Bandwidth in Hertz:** It is the range of frequencies contained in a composite signal or the range of frequencies a channel can pass. For example, let us consider the bandwidth of a subscriber telephone line as 4 kHz.
- **Bandwidth in Bits per Seconds:** It refers to the number of bits per second that a channel, a link, or rather a network can transmit. For example, we can say the bandwidth of a Fast Ethernet network is a maximum of 100 Mbps, which means that the network can send 100 Mbps of data.

Note: There exists an explicit relationship between the bandwidth in hertz and the bandwidth in bits per second. An increase in bandwidth in hertz means an increase in bandwidth in bits per

second. The relationship depends upon whether we have baseband transmission or transmission with modulation.

LATENCY

In a network, during the process of data communication, latency(also known as delay) is defined as the total time taken for a complete message to arrive at the destination, starting with the time when the first bit of the message is sent out from the source and ending with the time when the last bit of the message is delivered at the destination. The network connections where small delays occur are called “Low-Latency-Networks” and the network connections which suffer from long delays are known as “High-Latency-Networks”.

High latency leads to the creation of bottlenecks in any network communication. It stops the data from taking full advantage of the network pipe and conclusively decreases the bandwidth of the communicating network. The effect of the latency on a network’s bandwidth can be temporary or never-ending depending on the source of the delays. Latency is also known as a ping rate and is measured in milliseconds(ms).

- In simpler terms latency may be defined as the time required to successfully send a packet across a network.
- It is measured in many ways like a round trip, one-way, etc.
- It might be affected by any component in the chain utilized to vehiculate data, like workstations, WAN links, routers, LAN, and servers, and eventually may be limited for large networks, by the speed of light.

Latency = Propagation Time + Transmission Time + Queuing Time + Processing Delay

Propagation Time

It is the time required for a bit to travel from the source to the destination. Propagation time can be calculated as the ratio between the link length (distance) and the propagation speed over the communicating medium. For example, for an electric signal, propagation time is the time taken for the signal to travel through a wire.

Propagation time = Distance / Propagation speed

Example:

Input: What will be the propagation time when the distance between two points is 12, 000 km?

Assuming the propagation speed to be 2.4×10^8 m/s in cable.

Output: We can calculate the propagation time as-

$$\text{Propagation time} = (12000 * 10000) / (2.4 * 10^8) = 50 \text{ ms}$$

Transmission Time

Transmission Time is a time based on how long it takes to send the signal down the transmission line. It consists of time costs for an EM signal to propagate from one side to the other, or costs like the training signals that are usually put on the front of a packet by the sender, which helps the receiver synchronize clocks. The transmission time of a message relies upon the size of the message and the bandwidth of the channel.

$$\text{Transmission time} = \text{Message size} / \text{Bandwidth}$$

Example:

Input: What will be the propagation time and the transmission time for a 2.5-kbyte

message when the bandwidth of the network is 1 Gbps? Assuming the distance between sender and receiver is 12, 000 km and speed of light is $2.4 * 10^8$ m/s.

Output: We can calculate the propagation and transmission time as-

$$\text{Propagation time} = (12000 * 10000) / (2.4 * 10^8) = 50 \text{ ms}$$

$$\text{Transmission time} = (2560 * 8) / 10^9 = 0.020 \text{ ms}$$

Note: Since the message is short and the bandwidth is high, the dominant factor is the propagation time and not the transmission time(which can be ignored).

Queuing Time

Queuing time is a time based on how long the packet has to sit around in the router. Quite frequently the wire is busy, so we are not able to transmit a packet immediately. The queuing time is usually not a fixed factor, hence it changes with the load thrust in the network. In cases like these, the packet sits waiting, ready to go, in a queue. These delays are predominantly characterized by the measure of traffic on the system. The more the traffic, the more likely a packet is stuck in the queue, just sitting in the memory, waiting.

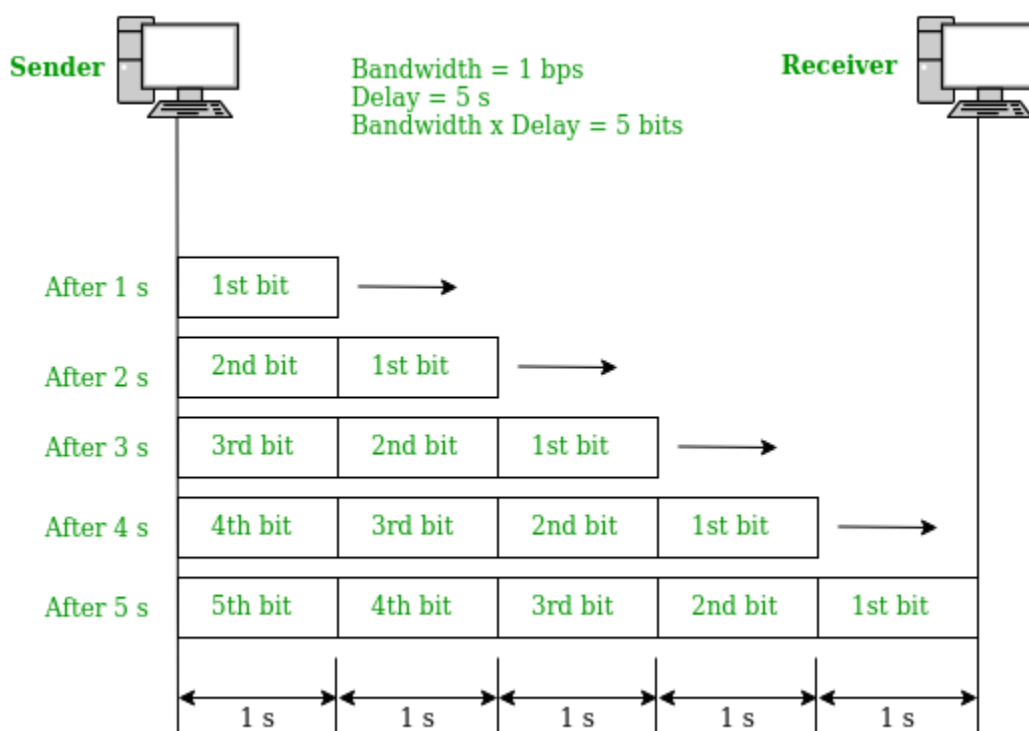
Processing Delay

Processing delay is the delay based on how long it takes the router to figure out where to send the packet. As soon as the router finds it out, it will queue the packet for transmission. These costs are predominantly based on the complexity of the protocol. The router must decipher enough of the packet to make sense of which queue to put the packet in. Typically the lower-level layers of the stack have simpler protocols. If a router does not know which physical port to send the packet to, it will send it to all the ports, queuing the packet in many queues immediately. Differently, at a higher level, like in IP protocols, the processing may include making an ARP request to find out the physical address of the destination before queuing the packet for transmission. This situation may also be considered as a processing delay.

BANDWIDTH – DELAY PRODUCT

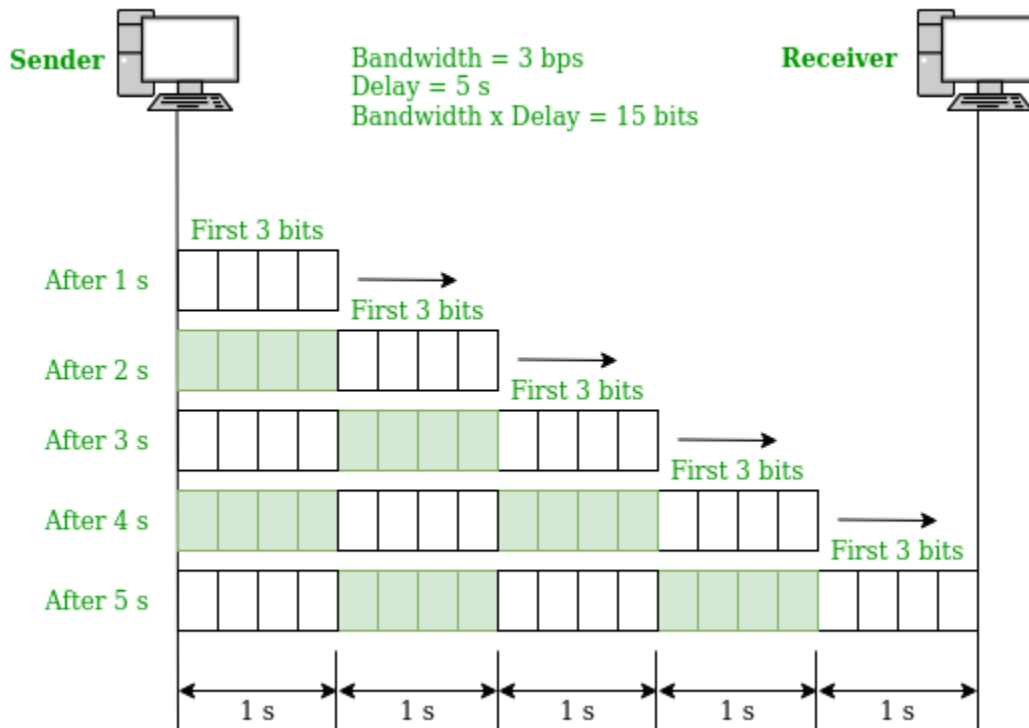
[Bandwidth and Delay](#) are two performance measurements of a link. However, what is significant in data communications is the product of the two, the bandwidth-delay product. Let us take two hypothetical cases as examples.

Case 1: Assume a link is of bandwidth 1bps and the delay of the link is 5s. Let us find the bandwidth-delay product in this case. From the image, we can say that this product 1×5 is the maximum number of bits that can fill the link. There can be close to 5 bits at any time on the link.



Bandwidth Delay Product

Case 2: Assume a link is of bandwidth 3bps. From the image, we can say that there can be a maximum of $3 \times 5 = 15$ bits on the line. The reason is that, at each second, there are 3 bits on the line and the duration of each bit is 0.33s.



Bandwidth Delay

For both examples, the product of bandwidth and delay is the number of bits that can fill the link. This estimation is significant in the event that we have to send data in bursts and wait for the acknowledgment of each burst before sending the following one. To utilize the maximum ability of the link, we have to make the size of our burst twice the product of bandwidth and delay. Also, we need to fill up the full-duplex channel. The sender ought to send a burst of data of $(2 \times \text{bandwidth} \times \text{delay})$ bits. The sender at that point waits for the receiver's acknowledgement for part of the burst before sending another burst. The amount: $2 \times \text{bandwidth} \times \text{delay}$ is the number of bits that can be in transition at any time.

THROUGHPUT

Throughput is the number of messages successfully transmitted per unit time. It is controlled by available bandwidth, the available signal-to-noise ratio, and hardware limitations. The maximum

throughput of a network may be consequently higher than the actual throughput achieved in everyday consumption. The terms 'throughput' and 'bandwidth' are often thought of as the same, yet they are different. Bandwidth is the potential measurement of a link, whereas throughput is an actual measurement of how fast we can send data.

Throughput is measured by tabulating the amount of data transferred between multiple locations during a specific period of time, usually resulting in the unit of bits per second(bps), which has evolved to bytes per second(Bps), kilobytes per second(KBps), megabytes per second(MBps) and gigabytes per second(Gbps). Throughput may be affected by numerous factors, such as the hindrance of the underlying analog physical medium, the available processing power of the system components, and end-user behavior. When numerous protocol expenses are taken into account, the use rate of the transferred data can be significantly lower than the maximum achievable throughput.

Let us consider: A highway that has a capacity of moving, say, 200 vehicles at a time. But at a random time, someone notices only, say, 150 vehicles moving through it due to some congestion on the road. As a result, the capacity is likely to be 200 vehicles per unit time and the throughput is 150 vehicles at a time.

Example:

Input: A network with bandwidth of 10 Mbps can pass only an average of 12, 000 frames per minute where each frame carries an average of 10, 000 bits. What will be the throughput for this network?

Output: We can calculate the throughput as-

$$\text{Throughput} = (12,000 \times 10,000) / 60 = 2 \text{ Mbps}$$

The throughput is nearly equal to one-fifth of the bandwidth in this case.

For more, refer to the [Difference between Bandwidth and Throughput](#).

JITTER

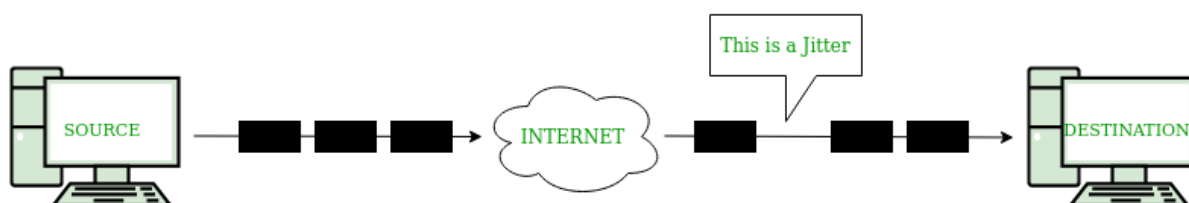
Jitter is another performance issue related to the delay. In technical terms, jitter is a "packet delay variance". It can simply mean that jitter is considered a problem when different packets of data face different delays in a network and the data at the receiver application is time-sensitive, i.e. audio or video data. Jitter is measured in milliseconds(ms). It is defined as an interference in the normal order of sending data packets. For example: if the delay for the first packet is 10 ms,

for the second is 35 ms, and for the third is 50 ms, then the real-time destination application that uses the packets experiences jitter.

Simply, a jitter is any deviation in or displacement of, the signal pulses in a high-frequency digital signal. The deviation can be in connection with the amplitude, the width of the signal pulse, or the phase timing. The major causes of jitter are electromagnetic interference(EMI) and crosstalk between signals. Jitter can lead to the flickering of a display screen, affects the capability of a processor in a desktop or server to proceed as expected, introduce clicks or other undesired impacts in audio signals, and loss of transmitted data between network devices.

Jitter is harmful and causes network congestion and packet loss.

- [Congestion](#) is like a traffic jam on the highway. Cars cannot move forward at a reasonable speed in a traffic jam. Like a traffic jam, in congestion, all the packets come to a junction at the same time. Nothing can get loaded.
- The second negative effect is [packet loss](#). When packets arrive at unexpected intervals, the receiving system is not able to process the information, which leads to missing information also called “packet loss”. This has negative effects on video viewing. If a video becomes pixelated and is skipping, the network is experiencing a jitter. The result of the jitter is packet loss. When you are playing a game online, the effect of packet loss can be that a player begins moving around on the screen randomly. Even worse, the game goes from one scene to the next, skipping over part of the gameplay.



Jitter

In the above image, it can be noticed that the time it takes for packets to be sent is not the same as the time in which they will arrive at the receiver side. One of the packets faces an unexpected delay on its way and is received after the expected time. This is jitter.

A jitter buffer can reduce the effects of jitter, either in a network, on a router or switch, or on a computer. The system at the destination receiving the network packets usually receives them from the buffer and not from the source system directly. Each packet is fed out of the buffer at a regular rate. Another approach to diminish jitter in case of multiple paths for traffic is to

selectively route traffic along the most stable paths or to always pick the path that can come closest to the targeted packet delivery rate.

Factors Affecting Network Performance

Below mentioned are the factors that affect the network performance.

- [Network Infrastrucutre](#)
- Applications used in the Network
- [Network Issues](#)
- [Network Security](#)

Network Infrastructure

Network Infrastructure is one of the factors that affect network performance. Network Infrastructure consists of routers, switches services of a network like IP Addressing, wireless protocols, etc., and these factors directly affect the performance of the network.

Applications Used in the Network

Applications that are used in the Network can also have an impact on the performance of the network as some applications that have poor performance can take large bandwidth, for more complicated applications, its maintenance is also important and therefore it impacts the performance of the network.

Network Issues

Network Issue is a factor in Network Performance as the flaws or loopholes in these issues can lead to many systemic issues. Hardware issues can also impact the performance of the network.

Network Security

Network Security provides privacy, data integrity, etc. Performance can be influenced by taking network [bandwidth](#) which has the work of managing the scanning of devices, encryption of data, etc. But these cases negatively influence the network.